

Linux – Important Notes

◆ What is Linux?

- Linux = open-source **kernel** (OS engine)
 - Ubuntu, Mint, Fedora → Linux-based operating systems
 - Linux-la **GUI + CLI** rendum irukum
 - Developers mostly **terminal** use pannuvaanga
 - Most servers Linux-la dhaan run aagum
 - So devs Linux basics therinjirukanum
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◆ Why we use Linux (for developers)

- Project run panna real environment
- Packages install panna
- Server connect panna
- Errors/logs check panna
- App deploy panna

👉 Not just compiling 👉 **Run + manage project**

◆ Basic Workflow

- Project folder open
 - Main file run
 - Terminal shows output or error
 - Error illa → success
 - Error vandha → fix → run again
 - home la poi paaru nu sonna we should enter /home/username we should not enter /home only because it will meaningless without the username
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◆ Key Points:

- cursor wont work in linux server / ubuntu

- meta data - size time created user group
- data-actual content

◆ **Creating and Editing Files:**

- pwd - present working directory (enga work panitu irukom nu kaatum)
- uname -a: -operating system version
- whoami - used to retrieve username
- clear / (ctrl +L) - used to clear the screen (if you want the latest commands used press up arrow button).
- history - full history of commands..
- mkdir- Create a new directory with given name(cmd-mkdir

new directory name).

- ls – List the content of directory (the contents are color of

blue its has a directory).

- vi textfilename (vi hello.txt(extension must)) - create a text file and enter in it...
- after entering the text file press i for inserting text mode after writing press escape button and click ':' wq for saving and exiting / :q! for exiting without saving
- sudo - run the command with administrator (root) permissiong - endha prechanaiyum illama run pandrathuku help aagum
- apt-get - package manager in Ubuntu for installing software
- install - tells the system that you want to install something
- vim - a sample package name - vim text editor
- nano - to create or edit a file....It has it build options in the bottom of the screen...for example we have to press ctrl+X to close the file and Ctrl is must for selecting the combinations
- Ctrl + Shift + C - to copy in Vi editor
- Ctrl + Shift + V - to paste in Vi editor
- we can use either vim or nano editor

- to create a dummy / empty file we should 'touch' filename..and also using touch file we can update the timestamp of a file and to create multiple files at a single shot.
- rm filename - to remove a file.....rm*.txt to remove all the file extensions with .txt and rm * to remove all the files in that directory..
- ls * to list all the files and ls *.txt to list all the text files.
- cat hello.txt - to list the file in the terminal itself (Not going to another screen)..
cat filename » newfilename - to copy the content of a filename and pasting it in a new file..
- mv - to move / rename a file...same purpose only to move the content of a file to another file
- echo - print in linuxecho "mithun" » hello.txt — meaning adding the word mithun in the file hello.txt it will add in the last
- rmdir- to remove a particular directory.

◆ File System Navigation (LS and CD) :

- pwd – shows current working directory
- ls – list files and folders in current directory
- ls -l – list files in long format, sorted by time,in reverse order andls -list files , -l - long listing format, -t - sort by time, -r - reverse order
- ls -l – detailed list of files and folders
- ls -a – shows hidden files also
- ls -la – detailed list including hidden files
- ls -lh – detailed list with readable file sizes
- ls foldername – list contents inside a specific folder
- ls /path/to/folder – view another folder without going there
- cd foldername – enter inside a directory
- cd .. – move one directory back
- cd ../.. – move two directories back
- cd – go to home directory

- `cd ~` – go to home directory
 - `cd /` – go to root directory
 - `cd -` – go to previous directory
 - `cd /path/to/folder` – go directly to a full path
 - while checking any file the extension is very important for example if i want to see a file inside demo then we have give `cd abc/123/demo test.txt`
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◆ **Hard links, Soft links, and Alias :**

- Hard link creation - `ln filename hardlinkfilename` (e.g., `ln original.txt hardlink.txt`)
 - Soft link creation - `ln -s filename softlinkfilename` (e.g., `ln -s original.txt softlink.txt`)
 - Hard links - Directly point to file data (same inode); original file delete pannalum hard link work aagum.
 - Soft links - Filename/path ah point pannum (shortcut maari); original file delete panna soft link broken aagum.
 - Hard link limitations - Directories ku create panna mudiyadhu; different filesystem cross panna mudiyadhu.
 - Soft link advantages - Directories ku create panna mudiyum; different filesystem la kuda work aagum.
 - Alias creation - `alias shortname='full command'` (e.g., `alias ll='ls -la'`)
 - View aliases - `alias` command use panna current aliases ellam kaatum.
 - Remove alias - `unalias aliasname`
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◆ **Background Jobs, nohup, and Process Monitoring :**

- Background jobs na terminal ah block pannama background la run aagura process. So same terminal la innum commands run panna mudiyum.
- Run command in background - Command last la `&` add pannunga Example: `python app.py &`
- View background jobs - `jobs` Current terminal la run aagura background jobs ellam kaatum.
- Bring job to foreground - `fg %jobnumber` Example: `fg %1`

- Send job to background - `bg %jobnumber`
 - Kill a job - `kill %jobnumber` Illati process ID use pannitu kill PID
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◆ **nohup (No Hang Up)**

- `nohup` command use panna, terminal close pannalum process continue aagum.
 - Purpose - Terminal close aana kuda process stop aagadhu.
 - Run with `nohup` `nohup` command & Example: `nohup python app.py &`
 - Output file - Default ah output `nohup.out` la save aagum.
 - Custom output `nohup` command `> output.log 2>&1 &`
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◆ **Process Monitoring**

- Process monitoring na system la run aagura processes ah check panni manage pannradhu (CPU, memory usage etc).
 - View running processes - `ps` Detailed view → `ps -ef`
 - Real-time monitor - `top` Live CPU & memory usage kaatum.
 - Improved monitor - `htop` (install pannirundha)
 - Find process by name - `ps -ef | grep processname`
 - Kill process by PID - `kill PID`
 - Force kill - `kill -9 PID`
 - Check process ID - `pidof processname`
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◆ **Disk Management**

Definition: Disk management na Linux system la storage devices & partitions ah view, mount, manage pannradhu.

- Check disk space - `df -h`
- Check folder size - `du -sh foldername`
- List disks & partitions - `lsblk`
- Mount a disk - `mount /dev/sdb1 /mnt`

- Unmount a disk - `umount /mnt`
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◆ Zip / Unzip

Definition: Zip use pannitu files ah .zip archive ah compress pannuvom; unzip use pannitu extract pannuvom.

- Create zip file - `zip file.zip filename`
 - Zip a folder - `zip -r file.zip foldername`
 - Unzip file - `unzip file.zip`
 - Unzip to folder - `unzip file.zip -d foldername`
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◆ Tar / Untar

Definition: tar use pannitu multiple files ah single archive file ah combine pannuvom; mostly backup & transfer ku use pannuvanga.

- Create tar archive - `tar -cvf file.tar foldername`
 - Extract tar - `tar -xvf file.tar`
 - Create tar.gz - `tar -czvf file.tar.gz foldername`
 - Extract tar.gz - `tar -xzvf file.tar.gz`
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◆ Word Count (wc)

Definition: wc use pannitu lines, words, characters count pannalam.

- Count lines - `wc -l filename`
 - Count words - `wc -w filename`
 - Count characters - `wc -c filename`
 - Full count - `wc filename`
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◆ Sort

Definition: sort use pannitu file lines ah alphabetical or numerical order la arrange pannalam.

- Sort alphabetically - `sort filename`

- Sort reverse - `sort -r filename`
 - Sort numerically - `sort -n filename`
 - Remove duplicates - `sort -u filename`
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◆ Head

Definition: head use pannitu file oda first few lines (default 10) paakalam.

- View first 10 lines - `head filename`
 - View first N lines - `head -n 5 filename`
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◆ Grep

Definition: grep use pannitu file la specific word/pattern search pannalam.

- Search a word - `grep "word" filename`
 - Ignore case - `grep -i "word" filename`
 - Show line numbers - `grep -n "word" filename`
 - Count matches - `grep -c "word" filename`
 - Search recursively - `grep -r "word" foldername`
 - Invert match - `grep -v "word" filename`
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◆ SSH (Secure Shell)

Definition: SSH use pannitu remote server ku secure ah connect pannuvom.

- Connect to remote server - `ssh username@hostname`
 - Connect using port - `ssh -p port username@hostname`
 - Login using key file - `ssh -i keyfile username@hostname`
 - Exit SSH session - `exit`
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◆ SCP (Secure Copy)

Definition: SCP use pannitu files ah local system & remote system kulla secure ah transfer pannuvom.

- Copy file to remote - `scp file.txt username@hostname:/path`
- Copy file from remote - `scp username@hostname:/path/file.txt .`

Copy folder recursively - `scp -r foldername username@hostname:/path`

◆ Find

Definition: find use pannitu directory hierarchy la files/folders search pannalam based on condition.

- Find by name - `find . -name "filename"`
 - Find file type - `find . -type f`
 - Find directory type - `find . -type d`
 - Find by extension - `find . -name "*.txt"`
 - Find and delete - `find . -name "file.txt" -delete`
 - Find by size - `find . -size +10M`
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◆ AWK

Definition: awk na powerful text processing tool; filter & format panna use pannuvom.

- Print specific column - `awk '{print $1}' filename`
 - Print multiple columns - `awk '{print $1, $3}' filename`
 - Search and print - `awk '/word/ {print}' filename`
 - Count lines - `awk 'END {print NR}' filename`
 - Use delimiter - `awk -F ";" '{print $1}' filename`
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◆ Change Mode (chmod)

Definition: chmod use pannitu file permissions (read, write, execute) change pannuvom.

- Numeric mode - `chmod 755 filename`
- Give execute permission - `chmod +x filename`
- Remove write permission - `chmod -w filename`

- Recursive change - `chmod -R 755 foldername`
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◆ Scheduling using Cron

Definition: Cron na time-based scheduler; specific time la command/script automatically run aagum.

- Edit crontab - `crontab -e`
 - View crontab - `crontab -l`
 - Remove crontab - `crontab -r`
 - Cron format - `* * * * *` command (minute hour day month weekday)
 - Example (daily 2 AM) - `0 2 * * * /path/to/script.sh`
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◆ Scheduling using Airflow

Definition: Apache Airflow na workflow management tool; complex data pipelines schedule & monitor panna use pannuvanga.

- Start webserver - `airflow webserver`
 - Start scheduler - `airflow scheduler`
 - List DAGs - `airflow dags list`
 - Trigger DAG manually - `airflow dags trigger dag_id`
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◆ Scheduling using Shell Script

Definition: Shell script use pannitu multiple Linux commands ah oru file la write pannitu automate pannuvom.

- Create script file - `nano script.sh`
 - Add shebang line - `#!/bin/bash`
 - Make executable - `chmod +x script.sh`
 - Run script - `./script.sh`
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◆ Shell Scripting

Definition: Shell scripting na Linux commands ah script file la write pannitu together execute pannradhu.

- Create script file - nano script.sh
- Add shebang line - #!/bin/bash
- Make executable - chmod +x script.sh
- Run script - ./script.sh
- Run without execute permission - bash script.sh
- Variables - name="John"
- Print variable - echo \$name
- User input - read name
- If condition -

if

[condition] then commandfi

- For loop -

for i in 1 2 3do echo \$idone
